

The Rhythm of the Mind: Exploring the Link Between Music Preferences and Mental Health Through Machine Learning

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Abstract—Music has been recognized for its ability to evoke emotions, trigger memories, and influence mood [1]. In this paper, we explore the potential of leveraging machine learning techniques to predict an individual's mental health status based on their music choices. This innovative approach not only holds promise for mental health status checking but also offers the possibility of categorizing songs as "preferred" for mental health purposes. In addition, we propose a chatbot that can interact with users to assess their playlists and collect additional data from the users in a natural way. By leveraging the power of AI, we seek to uncover novel insights at the intersection of music and psychological well-being.

Index Terms—Machine Learning, Chatbot, Music, Mental Health

I. INTRODUCTION

A. Relationship Between Music and Mental Health

For centuries, scholars and philosophers have been fascinated by the intricate connection between music and the human mind [2], [3]. Central to this exploration, researchers raised a series of questions. Is the music one listens to reflective of their mental health? Does playing a musical instrument influence mental well-being? Can music be utilized therapeutically to improve mental health? Music's ability to evoke emotions, trigger memories, and alter moods has long been recognized [1]. By delving into an individual's musical preferences, we may uncover hidden emotional states that traditional diagnostic methods could miss. Therefore, examining the connection between an individual's musical preferences and their mental state offers a unique opportunity to uncover emotions or states of mind that might remain concealed through conventional diagnostic methods.

B. Existing Work Using Data Science in Mental Health and Music

The intersection of data science, mental health, and music has gained increasing attention in recent years, with several studies leveraging large datasets and machine learning techniques to explore these connections. Research has demonstrated the potential of using streaming data from music platforms to analyze associations between listening habits

and personality traits [4]. Studies have applied natural language processing (NLP) to song lyrics to detect emotional content [5], while other approaches have utilized sentiment analysis to analyze users' emotional states based on their music preferences and lyrics [6]. Additionally, researchers have begun developing predictive models to assess mental health risks by analyzing social media interactions, incorporating music listening patterns as an indicator of emotional well-being [7]. While these efforts have explored the associations between music preferences and mental health, detailed musical features are rarely utilized. When they are included, the analysis is typically limited to conventional correlation studies [8], [9].

These early efforts provide a strong foundation for further exploration and demonstrated the value of data-driven methods in uncovering relationships between music and mental health. However, much of the existing work focuses on correlation study. There remains significant opportunity to integrate more advanced AI methodologies to develop personalized insights based on music preference. Our research builds upon this body of work, applying sophisticated machine learning techniques and leveraging big data to deepen our understanding of the complex relationship between music and mental health.

C. Leveraging Machine Learning to Explore Music's Impact on Mental Health

This research introduces using machine-learning methodology to dissect various components of songs to gauge an individual's emotional disposition. This approach underscores the intricate interplay between music and mental health across a spectrum of dimensions. Furthermore, the study's practical application extends to recommending suitable songs for individuals navigating emotional challenges. We also designed a chatbot that can interact with users to assess their psychological well-being, recommend songs, and collect additional data for further improving the machine learning algorithms. In a changing landscape where the assessment and support of mental health are of paramount importance, this study

presents a potential avenue for exploring and comprehending psychological well-being in new and meaningful ways.

D. Practical Applications and AI-Driven Mental Health Support

The practical applications of this AI-driven research are profound. By harnessing the power of machine learning and big data, we can recommend personalized music selections to individuals facing emotional challenges. Additionally, an AI-based chatbot has the potential to engage users in real-time to assess psychological well-being, recommend appropriate music, and collect user feedback to further refine the machine learning models. This interaction would allow for continuous learning and improvement, transforming music into a dynamic tool for mental health support. In an era where mental health is a global priority, this AI-powered research presents a groundbreaking avenue for understanding psychological well-being, offering scalable, data-driven insights that can be applied across clinical, therapeutic, and everyday settings.

II. METHOD

A. Data Set

In this study, we leveraged a publicly available data set [10]. It contained the data collected through a survey of 300 participants on psychological well-being and their music playlists. Figure 1 shows all the features collected. Participants' mental health was measured using a scale (0-5) developed by the Canadian Mental Health Association, including the ability to find joy in life, resilience, maintaining a balanced lifestyle, emotional adaptability, and self-actualization [11]. General demographic information such as gender, age, and hours of music listening per day, was also collected. Furthermore, music characteristics of their favorite songs were extracted through the Spotify API:

- **Energy:** Calculated using features such as dynamic range, perceived loudness, timbre, onset rate, and general entropy.
- **Danceability:** Measures how suitable a track is for dancing, based on a combination of tempo, rhythm stability, beat strength, and regularity.
- **Valence:** Represents the mental positiveness of the track.
- **Tempo:** The beats per minute (BPM) estimate of the track.
- **Popularity:** Total number of plays the track has received.
- **Instrumentalness:** Indicates whether the track contains vocals or is purely instrumental.
- **Acousticness:** Determines whether the track is acoustic or not.
- **Liveness:** Measures if there is an audience present in the recording.

B. Machine Learning-Based Prediction

Our study utilizes a machine-learning-based classification approach to predict mental health status by integrating both musical and demographic features. The primary objective of this model is to discern an individual's mental health status

based on the data they provide. For classification, we used a binary outcome where individuals are classified as either 'healthy' or exhibiting poor well-being. Respondents with a WHO (Five) Well-Being Index score above 13 are categorized as 'healthy,' while those scoring 13 or below are identified as having poor well-being. A score of 13 or lower can be an indicator of potential depression, providing a meaningful threshold for mental health testing and classification.

To implement this predictive model, we utilized the `Scikit-learn` machine learning library, a well-established toolkit in the field of data science and machine learning. Seven different machine learning algorithms were applied and rigorously evaluated to compare their predictive accuracy and performance. These algorithms included:

- 1) Logistic Regression,
- 2) Decision Trees,
- 3) Random Forest,
- 4) SGD Classifier,
- 5) k-Nearest Neighbors (kNN),
- 6) Adaptive Boosting,
- 7) Bagging Classifier.

The dataset was split into training and test sets with an 80%/20% ratio, ensuring a robust evaluation process. The 80% of the data allocated for training allowed the model to learn from a significant portion of the dataset, while the remaining 20% was reserved for testing and validating the model's predictive capabilities. This split ensures that the model can generalize well to unseen data, which is crucial for real-world applications.

Feature selection also played a key role in improving model performance. Both demographic variables (such as age, gender, and education) and music features (such as energy, valence, tempo, danceability, etc.) were included to provide a holistic view of each individual's profile. The combination of music and demographic data allows the model to capture both personal and behavioral factors that may influence mental health.

Performance metrics such as accuracy, precision, recall, and F1-score were used to evaluate and compare the models.

C. Chatbot Design

To further enhance user engagement and provide a more personalized experience, we designed a chatbot to interact with users. The primary function of this chatbot is to converse with users, assess their mental health status through interactive dialogue, and make personalized music recommendations based on their emotional needs. This conversational agent could serve as a user-friendly interface that bridges the gap between complex machine-learning algorithms and the end-user. This proposed chatbot will potentially make it easier for individuals to engage with the system.

The chatbot can be integrated with the Music Genie adaptor, which incorporates the machine-learning-based algorithm developed in our study. The Music Genie adaptor allows the chatbot to utilize the predictive model's output to generate

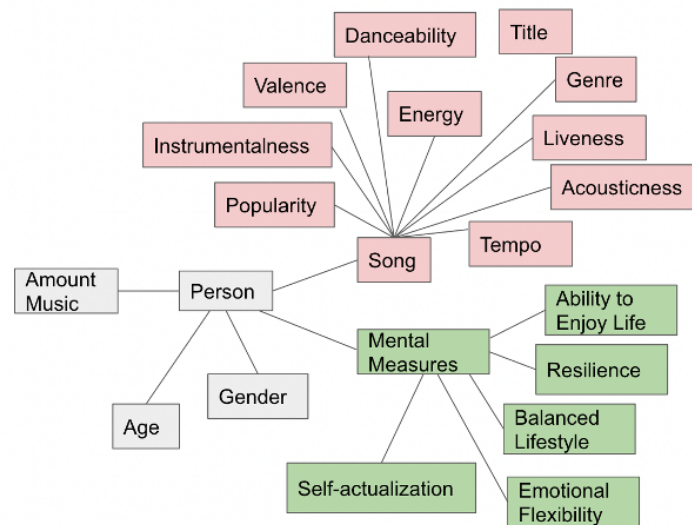


Fig. 1. Features Used in Training the Machine Learning Algorithms.

personalized music recommendations. By doing so, the chatbot could leverage both real-time user input and pre-trained models to make data-driven recommendations that are relevant to the user's emotional state.

In addition, the chatbot could interact with users by asking a series of questions related to their moods and music preferences. Through these conversations, data that can be fed into the machine-learning model can be gathered. We can leverage AI to generate appropriate music suggestions aimed at improving the user's well-being. The system could potentially be used to track changes in the user's mood over time, providing more accurate recommendations with continued use. In essence, the chatbot not only could assist users in finding suitable music but also could play a role in long-term mental health support by learning from each interaction.

Figure 2 shows the system overview.

III. RESULT

Among the machine learning classifiers we tested, Random Forest achieved the highest accuracy of 0.74, a precision of 0.73, and a recall of 0.98. (See Table 1 for a summary). Thus it was adopted for the rest of the study.

For the chatbot, we envision the initial functionality with a list of competency scenarios: (1) to collect a user's playlist to tell their psychological well-being using the machine algorithm, (2) to recommend suitable songs to the users, and (3) to conduct survey for additional data collection, which can further improve the machine learning performance through reinforcement learning. Figure 3 shows a mocked-up demo conversation between a user and the chatbot.

IV. DISCUSSION

In this study, we leveraged machine-learning methodologies to explore the connection between music preferences

and psychological well-being. By analyzing users' playlists and integrating demographic data, our approach offers an innovative way to independently predict mental health status. Music, as an integral part of human emotional expression, provides a unique window into individual states of mind, and our model capitalizes on this to forecast well-being through a data-driven lens. The integration of various machine learning algorithms allowed for comprehensive comparisons, revealing both the strengths and areas for improvement across different techniques.

In addition, this machine learning algorithm can be leveraged by chatbots to facilitate music therapy. A chatbot can potentially offer a seamless and interactive way to utilize AI for mental health applications, presenting music not just as a form of entertainment but as a therapeutic tool. A chatbot engages users in natural conversations, assessing their emotional state through dialogue and providing tailored music recommendations based on the machine learning model's predictions. Unlike traditional therapy or diagnostics, this tool can potentially offer a more personalized and immediate intervention, which allows users to engage with music as a therapeutic resource in real time. It is designed not only to provide immediate emotional relief but also to contribute to long-term mental health monitoring and improvement. Furthermore, the chatbot could collect valuable feedback from users, providing additional insights through surveys, which contribute to further refining the predictive model.

Despite the promising results, there are limitations. First, the size and diversity of the dataset used in this study could be expanded to improve the model's generalizability. Additionally, while the current design of the chatbot could provide a novel interface for user interaction, it mostly relies on pre-defined responses and suggestions. More dynamic, AI-driven conversations through generative AI could improve

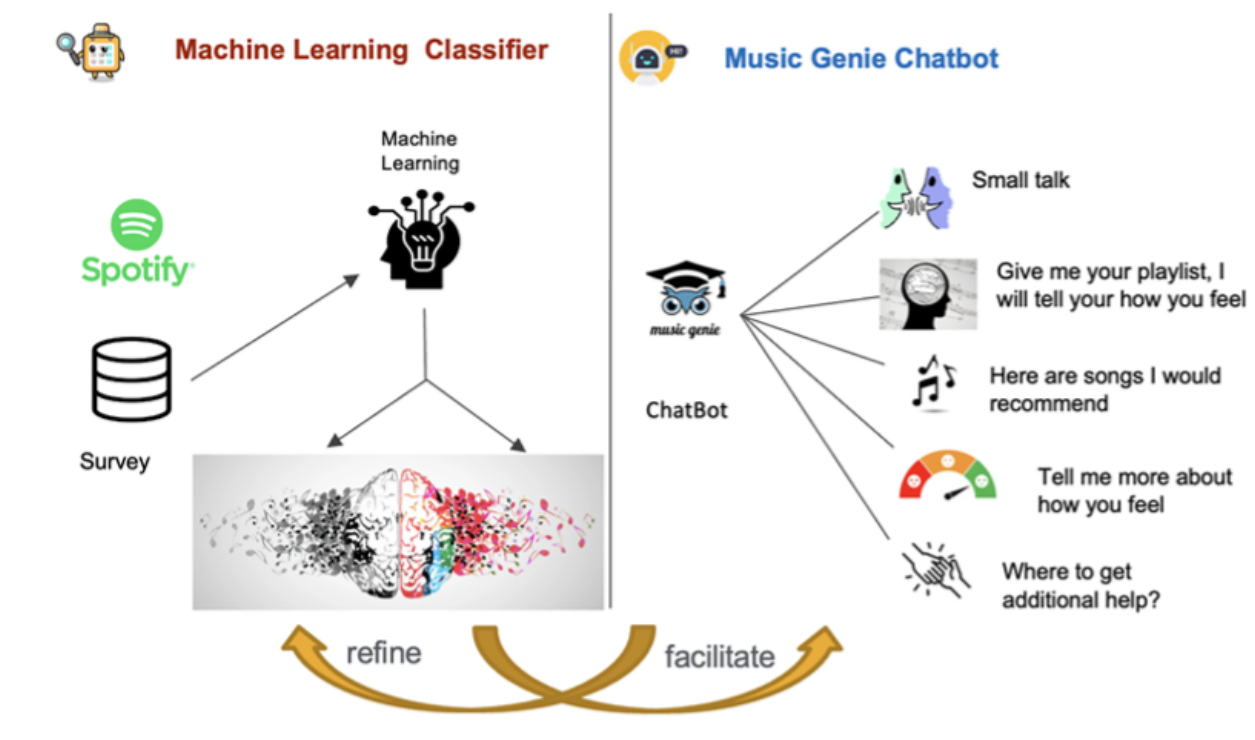


Fig. 2. System Overview.

Algorithm	Accuracy	Precision	Recall	F1 Score
SGD Classifier	66.37%	74.41%	87.07%	75.56%
Logistic Regression	70.09%	71.86%	85.71%	78.18%
Random Forest	73.62%	73.34%	98.57%	84.08%
Bagging Classifier	71.17%	73.41%	90.73%	81.09%
Decision Tree	62.07%	70.21%	80.49%	75.30%
Adaptive Boosting	65.52%	70.59%	87.80%	78.26%
K-nearest Neighbors	72.41%	72.73%	97.56%	83.33%

TABLE I
MACHINE LEARNING CLASSIFIER PERFORMANCE

user experience and the quality of data collected. In addition, although we discussed the potential impacts the proposed chatbot could bring to the users, we have not conducted user studies in terms of usability and efficacy. Furthermore, while the model focuses on binary classification (healthy vs. poor well-being), expanding the classification to reflect a broader range of mental health states could provide more detailed insights and interventions. Mental health is a multifaceted issue influenced by a wide range of biological, psychological, social, and environmental factors. Music represents just one dimension of mental well-being. Its impact can vary widely depending on an individual's context, preferences, and personal experiences.

V. CONCLUSION

In conclusion, our research presents a novel approach to mental health prediction by analyzing users' music preferences through machine learning. We successfully demonstrated that playlist data, combined with demographic information, can

be a useful indicator of an individual's mental health status. Our study provides a framework for further exploring the intersection of music, machine learning, and mental health, with potential applications in both personalized healthcare and broader mental health monitoring.

In addition, the AI-powered chatbot could offer a real-time, interactive tool that not only makes personalized music recommendations but also collects valuable user data. By fostering a conversational interface, a chatbot can bridge the gap between advanced machine learning models and user-friendly, accessible mental health interventions. It also demonstrates how AI can be integrated into therapeutic processes in a way that is adaptive, scalable, and efficient.

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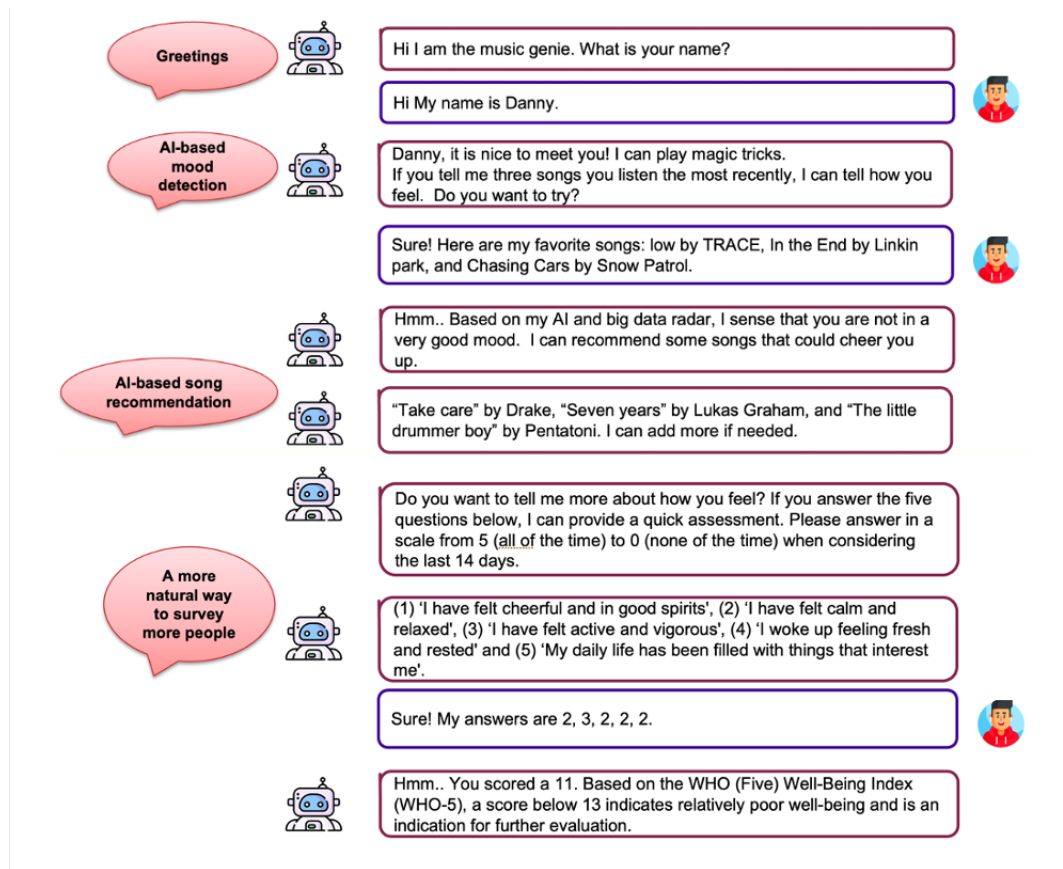


Fig. 3. Demonstration of a user conversation with the chatbot.

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